Atom-to-Reactor: Two-Dimensional Composition Grid, Solvation Models, and Fitting vs. Prediction

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This note specifies how the three design choices—water content, EC/DEC solvent ratio, and thermodynamic solvation vector ($\Delta E_{\mathrm{solv},i}$)—enter the thermodynamic and kinetic calculations, and explains how kinetic fitting loops uniformly over every solvation vector (computed or experimentally fitted) across the kinetic structures (`M1`, `M1-split`) and the 5×5 composition grid.

1 The Three Specification Axes

When exploring the two-dimensional composition space together with the solvation description, three quantities are specified:

1. **Water content (vol% $\mathbf{H_2O}$):**

$$\text{water} \in \{0,\ 0.5,\ 1.0,\ 1.5,\ 2.0\}\ \text{vol\%}$$

   Added to the solvent before TMSPA (5 vol% by default). It sets the initial concentration $[\mathrm{H_2O}]_0$ and slightly dilutes $[\mathrm{TMSPA}]_0$ and $[\mathrm{EC}]_0$, while leaving all standard-state reaction free energies $\Delta G^{\circ}_{\mathrm{rxn},j}$ unchanged.

2. **Solvent ratio (DEC volume fraction f_{DEC}):**

$$f_{\mathrm{DEC}} \in \{0,\ 0.25,\ 0.50,\ 0.75,\ 1.00\} \qquad (x^{\circ}_{\mathrm{EC}} = 1 - f_{\mathrm{DEC}})$$

   Here $f_{\mathrm{DEC}} = 0$ is neat EC and $f_{\mathrm{DEC}} = 1$ is neat DEC. The solvent ratio acts on *both* thermodynamics and kinetics:

   - It mixes the neat-EC and neat-DEC solvation energies $\Delta E_{\mathrm{solv},i}$ by solvent mole fraction $(x_{\mathrm{EC}}, x_{\mathrm{DEC}})$, shifting $\Delta G^{\circ}_{\mathrm{rxn},j}$, $K_{\mathrm{eq},j}$, and every Marcus barrier $\Delta G^{\ddagger}_{f,j}$.
   - It scales the initial ethylene carbonate concentration $[\mathrm{EC}]_0$ available for solvent attack (R8–R9), from $[\mathrm{EC}]_0 \approx 14.2$ M at $f_{\mathrm{DEC}} = 0$ (with 5 vol% TMSPA and 2 vol% $\mathrm{H_2O}$) through 7.1 M at $f_{\mathrm{DEC}} = 0.50$, down to 0 M at $f_{\mathrm{DEC}} = 1$.

3. **Thermodynamic solvation vectors (`thermo_models`, matching `solvset.py`):** Solvation inputs are organized in a two-level hierarchy of families (`thermo_models`) and concrete instances (snapshots or equilibrium fits), with gas-phase $\Delta G^{\circ}_{\mathrm{gas},i}$ from ωB97M-V fixed across all of them:

$$\begin{aligned}
\texttt{thermo_models} &= (\texttt{MD},\ \texttt{COSMO},\ \texttt{Experimental}) \\
\texttt{MD} &= [\texttt{snapshot_1}\ (\texttt{2026-10-08}),\ \texttt{snapshot_2}\ (\texttt{2026-11-12}),\ \ldots] \\
\texttt{COSMO} &= [\texttt{cosmors},\ \ldots] \\
\texttt{Experimental} &= [\texttt{exp_1}\ (\texttt{thermo2_2pctH2O}),\ \texttt{exp_2}\ (\texttt{thermo4_2pct_cosmors}),\ \ldots]
\end{aligned}$$

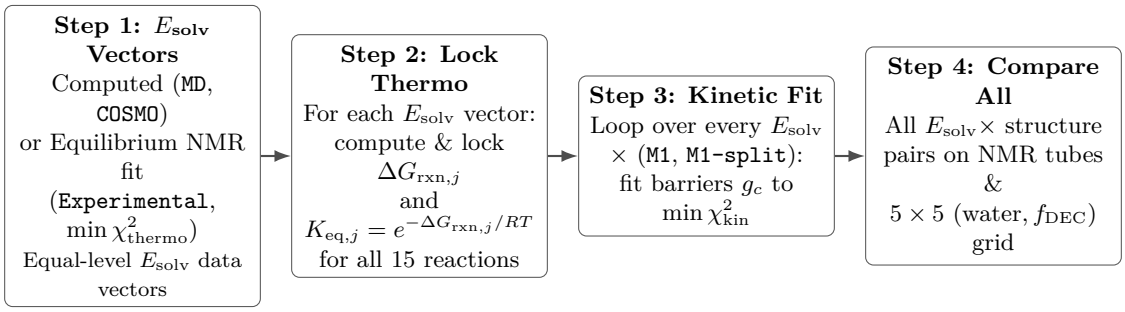
   Once the `Experimental` fits have been obtained from equilibrium NMR spectra, every `(thermo_model, instance)` pair resolves to a species solvation vector $E_{\mathrm{solv},i}$ **on the exact same footing**. Computed vectors (`MD`, `COSMO`) and fitted vectors (`Experimental`) are interchangeable E_{solv} data vectors for all downstream kinetic fits and grid evaluations.

`thermo_model`	Instance (`snapshot/fit`)	Source of $E_{\mathbf{solv}}$ vector	Description
`MD`	`snapshot_1` (`2026-10-08`)	Pure MACE MD	Computed MD solvation vector from the initial snapshot ($\delta = 0$)
`MD`	`snapshot_2` (`2026-11-12`)	Pure MACE MD	Computed MD solvation vector from the updated campaign ($\delta = 0$)
`COSMO`	`cosmors`	Pure openCOSMO-RS	Computed continuum solvation free energy $\Delta G_{\mathrm{solv},i}$ ($\delta = 0$)
`Experimental`	`exp_1` (`thermo2_2pctH2O`)	`2026-10-08` MD + δ	MD vector shifted within $\pm\xi\sigma$ by equilibrium 2% H_2O (EC:DEC 1 : 1) fit
`Experimental`	`exp_2` (`thermo4_2pct_cosmors`)	`COSMO` + δ	COSMO vector shifted within $\pm\xi\sigma$ by equilibrium 2% H_2O (EC:DEC 1 : 1) fit

Assumption when extrapolating `Experimental` vectors across $f_{\mathbf{DEC}}$
The species shifts $\delta E_{\mathrm{solv},i}$ in `exp_1` and `exp_2` were fitted at $f_{\mathrm{DEC}} = 0.50$ (EC:DEC 1 : 1). When evaluated at any other f_{DEC}, the underlying computed medium mixes the neat-EC and neat-DEC values while the shifts $\delta E_{\mathrm{solv},i}$ are carried over unchanged as a constant species offset.

2 Sequential Pipeline and All-vs-All Comparison Matrix

In the standard workflow, thermodynamics and kinetics are **not** fitted simultaneously. Equilibrium NMR data produces `Experimental` E_{solv} vectors that sit on the exact same level as computed `MD` and `COSMO` vectors. Downstream kinetic fitting then runs identically over *every* E_{solv} vector and *every* kinetic structure so that everything can be compared against everything:



1. **Step 1 — Assemble equal-level $E_{\mathbf{solv}}$ vectors:** Each `(thermo_model, instance)` yields a fixed species solvation vector $E_{\mathrm{solv},i}$:
   - For `MD` and `COSMO`, $E_{\mathrm{solv},i}$ is taken directly from the quantum/MD calculation (no equilibrium fitting).
   - For `Experimental`, equilibrium NMR points only are used to fit $\delta E_{\mathrm{solv},i}$ inside $\pm\xi\,\sigma_{\mathrm{solv},i}$ minimizing χ^2_{thermo}; once fitted, the resulting $E_{\mathrm{solv},i} = E^{\mathrm{base}}_{\mathrm{solv},i} + \delta E_{\mathrm{solv},i}$ becomes a fixed E_{solv} data vector at the exact same level as `MD` and `COSMO`.
2. **Step 2 — Lock $\Delta G_{\mathbf{rxn},j}$ and $K_{\mathbf{eq},j}$ for each vector:** From each E_{solv} vector and fixed gas-phase $\Delta G^{\circ}_{\mathrm{gas},i}$, compute and lock $\Delta G_{\mathrm{rxn},j}$ and $K_{\mathrm{eq},j} = \exp(-\Delta G_{\mathrm{rxn},j}/RT)$ for all 15 reactions before any kinetic fitting.
3. **Step 3 — Kinetic fit across all vectors and structures:** Holding $\Delta G_{\mathrm{rxn},j}$ and $K_{\mathrm{eq},j}$ fixed,

evaluate Marcus barriers and Eyring rate constants:

$$\Delta G^{\ddagger}_{f,j} = g_c \left(1 + \frac{\Delta G_{\mathrm{rxn},j}}{4g_c}\right)^2, \qquad k_{f,j}(T) = \frac{k_B T}{h} \exp\left(-\frac{\Delta G^{\ddagger}_{f,j}}{k_B T}\right), \qquad k_{r,j}(T) = \frac{k_{f,j}(T)}{K_{\mathrm{eq},j}(T)},$$

and fit *only* g_c to minimize χ^2_{kin} for **every** `(thermo_model, instance)` and **every** kinetic structure:

- `M1`: 4 barriers ($g_{\mathrm{hydrolysis}}$ shared by R1–R3, N12, N15; plus $g_{\mathrm{transfer}}, g_{\mathrm{condensation}}, g_{\mathrm{solvent_attack}}$).
- `M1-split`: 5 barriers (hydrolysis split into g_{R1} and g_{R23}) or 6 barriers (with g_{N15} separate as well).

With the 5 current E_{solv} vectors and 2 kinetic structures, this produces a $5 \times 2 = 10$-cell matrix of kinetic parameterizations (growing cleanly whenever a new snapshot or `thermo_fit` is added).

4. **Step 4 — All-vs-all comparison on NMR tubes and 5×5 grid:** Every `(thermo_model, instance, kinetic_structure)` cell can now be compared directly against every other cell—both in fit quality (χ^2, residual z-scores, barriers g_c) on measured tubes and in forward predictions ($C_{\mathrm{eq}}, C(t)$) over all 25 (water, f_{DEC}) compositions ($10 \times 25 = 250$ grid runs for the current 5 vectors).

Exception: the 12/11/2026 neat-DEC tube (`M1-split-dG`)

Whenever equilibrium NMR points exist (as in EC:DEC 1 : 1), `Experimental` vectors are fitted strictly in Step 1 and *only* g_c is fitted in Step 3. The 12/11/2026 neat-DEC tube was recorded only from 1 to 9 min and has no equilibrium spectrum yet, so no Step-1 `Experimental` vector exists in neat DEC. Freeing $\Delta G_{\mathrm{R1}}, \Delta G_{\mathrm{R2}}$ within their MD covariance Σ_{solv} during the kinetic fit (`M1-split-dG`) was a temporary workaround specifically for that un-equilibrated tube.

Pipeline step	**Fitted quantities**	**Fixed inputs**	**Role in all-vs-all comparison matrix**
Step 1: Build E_{solv} vectors	$\delta E_{\mathrm{solv},i}$ in $\pm\xi\,\sigma_{\mathrm{solv},i}$ (`Experimental` only)	$\Delta G^{\circ}_{\mathrm{gas},i}$, base $E_{\mathrm{solv},i}$, $\xi = 1$	Produces equal-level E_{solv} vectors across `MD`, `COSMO`, and `Experimental` (equilibrium NMR only).
Step 2: Thermo lock	None	Each E_{solv} vector + $\Delta G^{\circ}_{\mathrm{gas},i}$	Computes and locks $\Delta G_{\mathrm{rxn},j}$ and $K_{\mathrm{eq},j}$ for every `(thermo_model, instance)`.
Step 3: Kinetic fit	Barriers g_c only (4 in `M1`; 5–6 in `M1-split`)	Locked $\Delta G_{\mathrm{rxn},j}, K_{\mathrm{eq},j}$ from Step 2	Loops over every E_{solv} vector × structure on reacting NMR points to minimize χ^2_{kin}. (`M1-split-dG` is the neat-DEC exception.)
Step 4: Compare all	None (forward run)	E_{solv} vector + g_c + (water, f_{DEC})	All-vs-all matrix: $N_{E_{\mathrm{solv}}} \times 2$ structures ($5 \times 2 = 10$ models $\times 25$ grid points = 250 simulations).